

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Errors in the Data Stream!



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can explain the concept of error correction.
- I can describe how data and error correction are related.
- I can identify error correction in transmitting data.

Ask your partner:

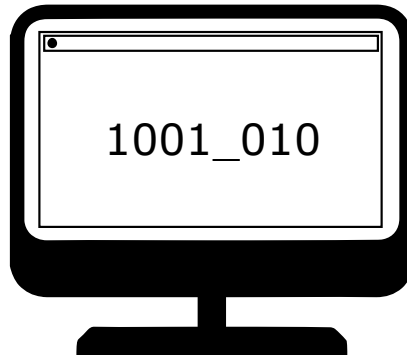
- What kind of strategy do you use to find mistakes in a large list of elements?
- In your opinion, what are the most common problems when transmitting data?





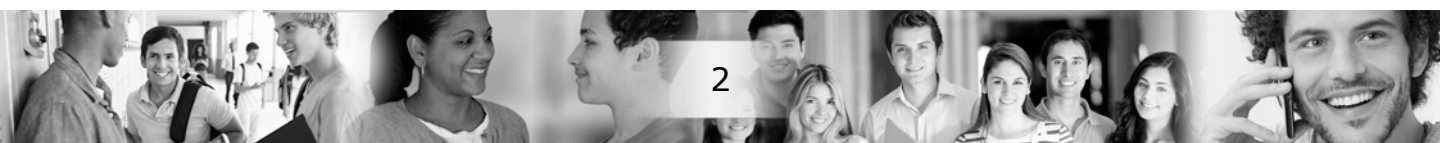
Let's Explore!

Mathew and Ann are talking about error correction schemas and their uses. Let's identify some of them



Instructions: Read the text below.

- Mathew:** Ok, Ann. We are about to finish programming the company's web-based software, but we must analyze some of the variables that are there so that we can choose the error correction method the program needs to protect the information while it is exchanging data.
- Ann:** You are right; remember that when we transmit data, errors are present in different ways during the process. For example, when the program is loading information from the web to store it in the cache of the computer; if there is a voltage drop, the electrical noise in the line can cause spurious bits or a bit flipping in the data stream. As a result, data errors in the disk occur.
- Mathew:** Of course, the simple coding of the program allows some wrong characters, but there are cases in which a large number of wrong characters become a problem for the program because they produce bugs and failures. That's why engineers in the company are developing techniques to ensure the correct transmission of data. But, is there any way to ensure that process?
- Ann:** Sure, there are some schemes, such as the parity one. There, programmers add an extra bit to a set of bytes of data so that the extra bit joins to one bit when the number is odd. When there is an error, one of the bits disappears, and the extra bit joins to fix the error.
- Mathew:** I know our programmers are implementing the checksum schema to correct errors, because it adds the binary value of each block of data, and sends them along with the block of information. Finally, when it reaches its destination, the program checks the data to see if they match.
- Ann:** Sure, but personally, I'm using the Cyclical Redundancy Check method to avoid and correct errors, as it sends data to the program and divides the blocks into specific numbers, when the data reaches its destination the program joins and calculates them again.
- Mathew:** Do not forget we can also implement the Hamming Code to provide high reliability, as it also corrects errors automatically. This scheme encrypts data and bits together to generate a code word in the file to ensure that the file at its destination only opens with the same code word.





Let's Practice!

Mathew is explaining to us what the causes of errors in data are

Instructions: Underline the sentences that use the infinitive of purpose.

When programs are transmitting data at high speeds to store it in the pc, disk, or cloud-based app, the information becomes sensitive to possible errors. To avoid such problems, software developers are implementing different error handling tools, such as the checksum schema or the Hamming code schema. Such schemas help the program to encrypt the information during the transmission process.



Let's Practice!

Explaining what error correction is about

Instructions: Read the text and write the missing words. Use: voltage drops, the correct transmission, correct data errors.

1. Engineers are planning to _____ through error correction techniques.
2. When programs transfer data, external factors such as _____ can generate errors in the data.
3. In order to prevent such errors, software developers implement error correction schemas to guarantee _____ of data.



Extension Activity

It is time for you to show how much you know about error correction tools.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Do more research on error correction in data transmission.
- Create a poster with the main characteristics of error correction in data transmission.
- Use present continuous and the infinitive of purpose to express your ideas.
- Explain the characteristics of the error correction schemas to your partners.

Self-Assessment

Yes

No

- I do extra research on error correction in data transmission.
- I create a poster with the main characteristics of error correction in data transmission using the grammar of the unit.
- I share my poster and explain it to my partners.

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Target Words

- **flip** (verb): to move quickly
e.g. If there is a voltage drop, the electrical noise in the line can cause spurious bits or a bit flipping in the data stream.
- **block** (noun): a group of data together
e.g. It adds the binary value of each block of data, and sends them along with the block of information.
- **encrypt** (verb): to convert or decode
e.g. This scheme encrypts data and bits together to generate a code word in the file to ensure that the file at its destination only opens with the same code word.
- **transmit** (verb): to communicate or to transfer information
e.g. You are right; remember that when we transmit data, errors are present in different ways during the process.
- **reliability** (noun): the degree in which something is trusted because it works well
e.g. Do not forget we can also implement the Hamming Code to provide high reliability, as it also corrects errors automatically.



Learning Points

Infinitives of Purpose with Present Continuous to Talk about the Purpose of Plans

We use the **infinitive of purpose** to express the reason why someone or something performs an activity.

As you probably know by now, we use the **present continuous** form to describe those activities that we are performing at the moment of speaking (ongoing actions) or to describe any action that is taking place.

However, if you want to express the reason why something is happening in either an ongoing action or a general action, you must use the infinitive of purpose.

The infinitive of purpose joins two ideas into one sentence, in which the first idea develops an action, and the second idea explains the reason of the first one to happen. In order to make a sentence using the **present continuous** form and **the infinitive of purpose**, you have to use the following rule:

Idea 1	Idea 2
Subject + to be + verb ing + complement	To + verb base form + complement
<i>Jane is analyzing the error solution software.</i>	<i>to provide precise tools to handle errors.</i>

Examples:

- The company **is implementing** different schemas **to analyze** the possible errors in the program.
- Jane **is talking** to the IT manager **to find** a prompt solution to the error handling problem they have.

Examples

- When we transmit data, errors are present in different ways during the process. For example, when the program is loading information from the web to store it in the cache of the computer.
- That's why engineers in the company are developing techniques to ensure the correct transmission of data.
- Our programmers are implementing the checksum schema to correct errors.
- Our programmers are implementing the checksum schema to correct errors.



Learning Points

Countable and Uncountable Nouns

It is important to differentiate what **countable** and **uncountable nouns** are.

When we talk about nouns, it is important to make the difference between **countable** and **uncountable** nouns.

COUNTABLE NOUNS

Use **countable nouns** to make reference to those individuals such as people, things, and animals that you can count, that is, those nouns you can pluralize by adding -s or -es at the end of the word.

Examples:

Program – programs

Disk – Disks

Process – Processes

- *The IT team is running some error correction **programs** int the **system** to ensure the quality of the service.*
- *There are some **processes** we need to follow in order guarantee the **program**.*

UNCOUNTABLE NOUNS

Use uncountable nouns to refer to those things that one can perceive as a whole or a mass rather than an individual.

It is important to remember that **uncountable nouns** do not have plural form; therefore, you always use them in their singular form.

Examples:

Software

Hardware

Information

Data*

Note that the word **data** is the plural form of **datum**, but people commonly use data with either singular or plural verbs (*data **is** / data **are***).

Examples:

- *There is some **data** missing in the **software**.*
- *This is a piece of **software** you can use to verify the **data** in the system.*

Remember that nouns can work as either the subject or object of a sentence. Therefore, you can use countable and uncountable nouns as the object or subject of the ideas you create.





Learning Points

Examples:

- Error discovery software **is** important for our company, that's why we use this software.
- Programs **are** important for our company, that's why we use this piece of.

Notice how the words in bold work as the subject and object of the sentences.

Examples

- *Remember that when we transmit data, errors are present in different ways during the process.*
- *The simple coding of the program allows some wrong characters, but there are cases in which a large number of wrong characters become a problem for the program.*